

Weijia Zhang

2172003915 | zhangwj.charlie@gmail.com | [linkedin.com/in/weijia-charlie-zhang/](https://www.linkedin.com/in/weijia-charlie-zhang/) | charliedreemur.github.io | [Google Scholar](#)

EDUCATION

Yale University

M.S. in Computer Science (Thesis Track with Full Scholarship)

University of Illinois Urbana-Champaign (UIUC)

B.S in Computer Science and Mathematics (C.W. Gear Outstanding Undergraduate Student - One of two annual recipients)

Sep 2026 - May 2028

GPA: 4.0/4.0

Aug 2022 - May 2026

GPA: 3.7/4.0 (Dean's List)

WORK EXPERIENCE

TikTok

Research Scientist Intern, Multimodal LLMs/Agents

Jun. 2026 – Present

- Designed a self-evolving content-moderation agent framework centered on a ReAct agent that autonomously analyzes individual videos using specialized tools for frame sampling, ASR/OCR, metadata extraction, originality scoring, and watermark/celebrity detection; integrates multimodal evidence with policy-grounded reasoning to produce structured moderation decisions.
- Built a self-evolution loop where a *Reflector* aggregates Solver failures into root-cause attribution reports, and an *Evolver* add/revise/merge case-relevant skills to refine the Solver and improve 15.5% overall f1 score

Microsoft

Research Scientist Intern

July 2025 – Sep 2025

- Conducted LLM research for Microsoft 365 Excel Copilot, focusing on spreadsheet text-analysis And Office.js code generation.
- Developed the **TextAnalysisSFT** data pipeline for generating SFT data for Excel Copilot's TextAnalysis API: mined 2,000+ real-world examples from Kaggle, filtered text-heavy spreadsheets, generated user queries and Office.js programs, and validated executions with SheetEngine; the resulting dataset improved Office Scripts code-generation accuracy by 15%.

Reborn Network

AI Engineer

May 2023 – July 2023

- Developed a **role-playing agent Unity VR Game** enabling agents interaction through text/voice/VR actions in real time (< 1s latency)
- Introduced RAG/vector database to strengthen long-term agent memory, improving dialogue coherence score from 2/5 to 4/5
- Designed a reusable "character card" framework, enabling UGC ecosystem and reducing character persona configuration time by 300%

Tencent

Software Engineer, Wechat Group

Aug 2024 – Sep 2024

- Developed a cross-platform (Android & iOS) hardware performance analysis tool for WeChat Mini Programs, supporting 200+ partner teams in identifying performance bottlenecks
- Built Unity Mono Memory Profiler that discovered 40+ hidden memory allocation points, reducing memory-leak-related crash rate by 120%

PUBLICATIONS(100+ CITATION)

Weijia Zhang et al. "CUADebug: Diagnosing and Repairing Computer-Use Agent Failures" *EMNLP 2026*.

Weijia Zhang et al. "Every Act Has Its Price: Compressed Moral Composition in Frontier LLMs." *EMNLP 2026*.

Weijia Zhang et al. "How Much Vision Does Multimodal Reasoning Need?" *Under review at NeurIPS 2026*.

Weijia Zhang*, Zijia Liu*, Haoru Li*, et al. "SeeingEye: Agentic Information Flow Unlocks Multimodal Reasoning in Text-only LLMs." *arXiv:2510.25092; under review at AAAI 2026*.

Miaomiao Li et al. (including **Weijia Zhang**). "Understanding and Mitigating the Bias Inheritance in LLM-based Data Augmentation on Downstream Tasks." *ACL 2026*. **Oral**.

Gordon Dai* and **Weijia Zhang*** et al. "Artificial Leviathan: Exploring Social Evolution of LLM Agents Through the Lens of Hobbesian Social Contract Theory." *AI Behavioral Science Workshop, ACM SIGKDD 2024*.

Kunlun Zhu et al. (including **Weijia Zhang**). "Where LLM Agents Fail and How They Can Learn From Failures." *arXiv:2509.25370. Failure Modes in Agentic AI, ICML 2026*

PROJECT EXPERIENCE

OpenManus & OpenManus-RL (Researcher, GitHub 60k+ Stars) | [GitHub](#)

Jan 2025 – Jul 2025

- Collaborated with the official OpenManus team as a core contributor to OpenManus-RL, the follow-up project of OpenManus; the open-source ecosystem has accumulated 60,000+ GitHub stars
- Implemented a ReAct-style reasoning architecture (*Think* → *Act* → *Tool-use*) and multi-agent collaboration workflows; built an MCP-based tool-library framework for automated tool invocation and output processing
- Constructed high-quality SFT datasets for agent benchmarks including AlfWorld and WebShop; designed step-level reward signals to improve agent planning and memory capabilities

CUAAGentDebugger (First Author / Lead)

Jan 2026 – Present

- Designed a GUI-agent error taxonomy with 4 major categories and 29 subtypes, covering perception, interaction localization, task reasoning, and external system failures across mainstream OSWorld/CUA scenarios
- Built a self-evolving VLM-agent debugging framework that automatically identifies root causes from failed trajectories, distills them into reusable debugging skills, and enables agents to learn autonomously from historical failures
- Designed a dual-layer memory architecture with episodic and semantic memory, together with an intent-aware RAG retrieval mechanism that injects skills from similar-intent trajectories to improve diagnosis and re-rollout accuracy

Understanding and Mitigating Bias Inheritance in LLM-based Data Augmentation

Nov 2024 – Feb 2025

Researcher

- Executed 50+ fine-tuning experiments using Llama-3.1 and ChatGPT-o1-mini with controlled bias blend ratios
- Developed automated pipeline for synthetic data generation, bias injection, LoRA fine-tuning, and evaluation

SKILLS

Programming Languages: Python, C++, C#

Frameworks & Libraries: VERL, VLLM, LangGraph, LangChain, PyTorch, Tensorflow

AI Focus: SFT, RL/Post-training, LLM Agents, Multimodal Reasoning, RAG, Self-Evolving Agent, MAS,